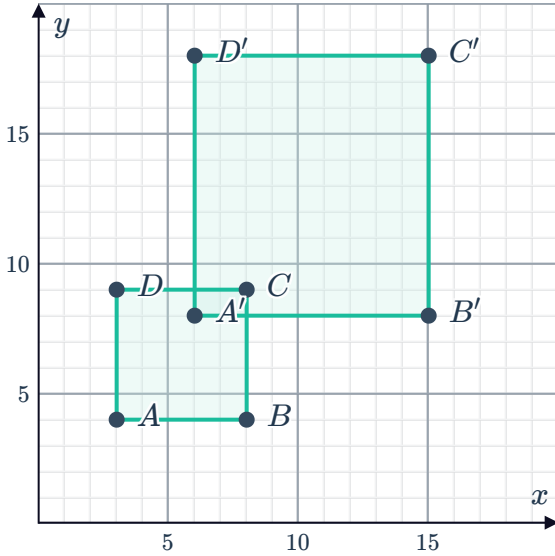


Dilations

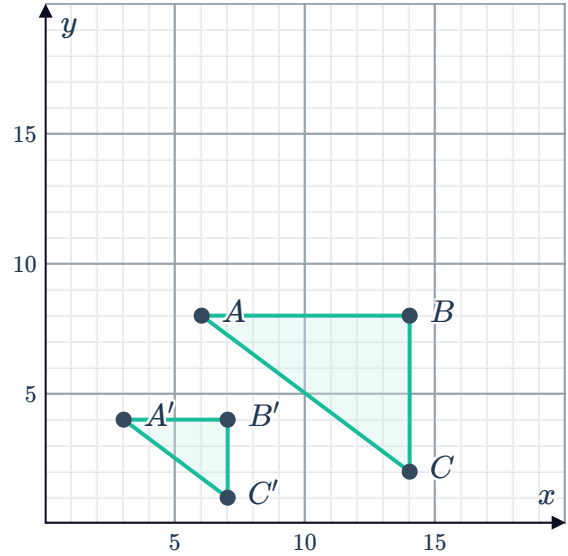


1 Determine whether the following graphs show a dilation. If yes, find the scale factor.

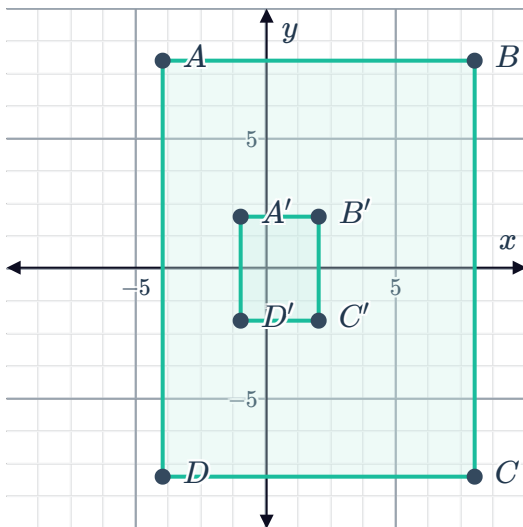
a



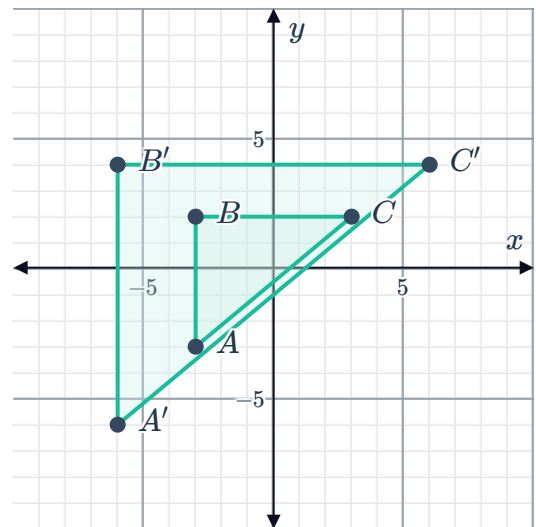
b



c



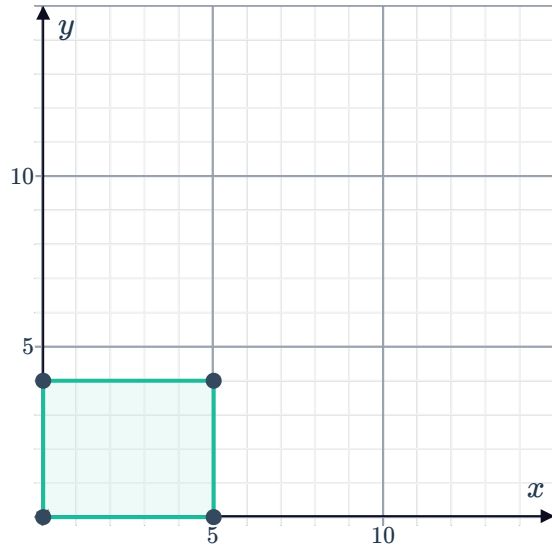
d



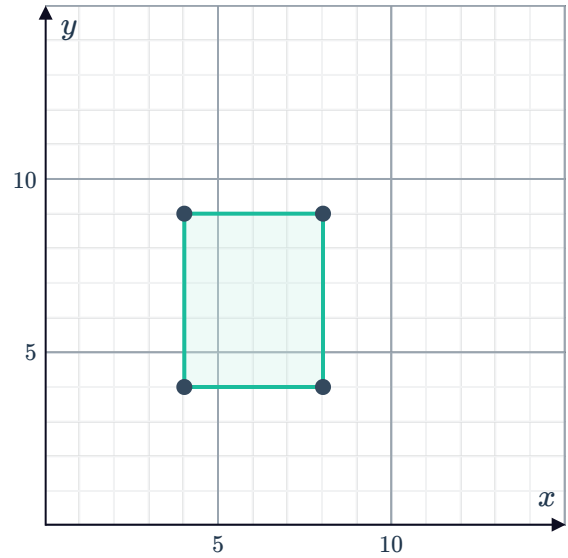
2 Dilate each figure using the given scale factor and the origin as the center of dilation.



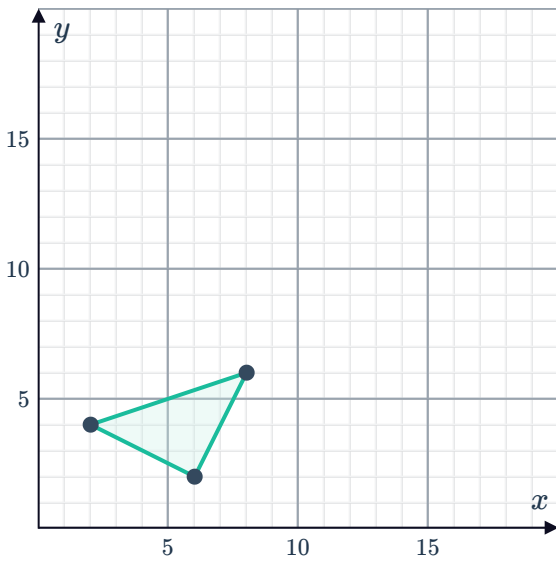
a Scale factor: 4



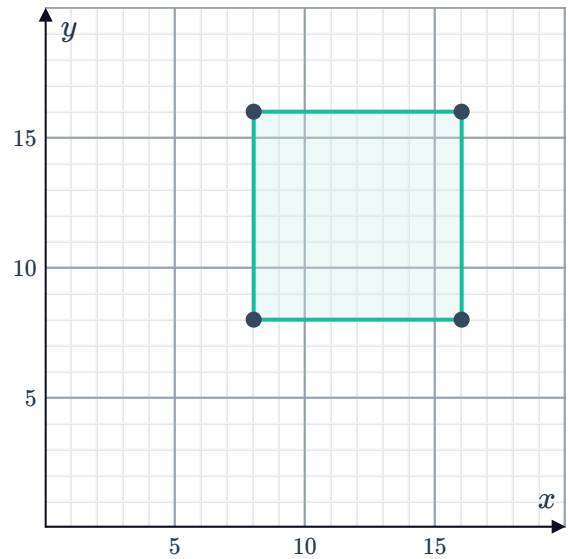
b Scale factor: 2



c Scale factor: 1.5



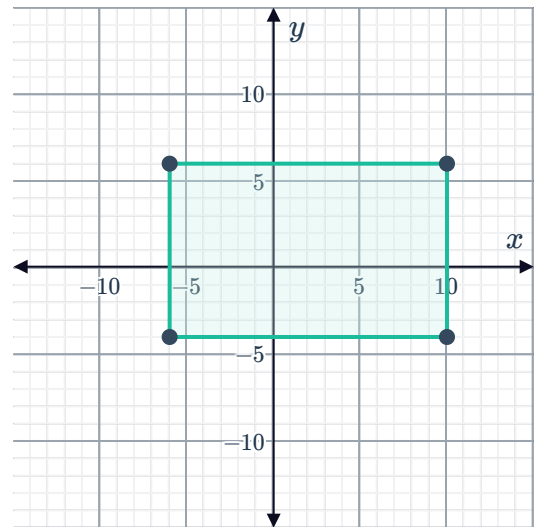
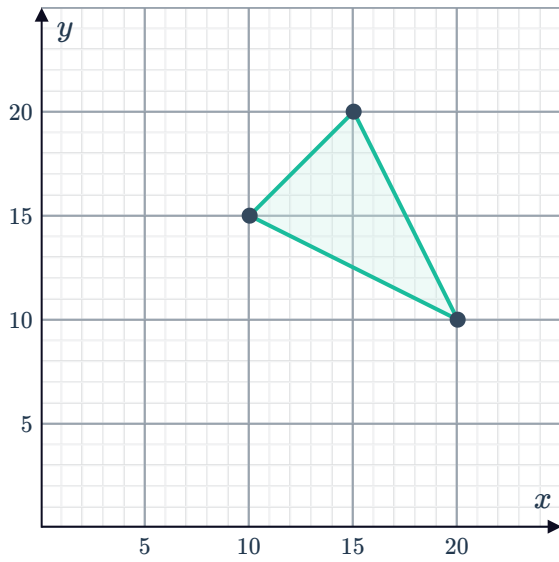
d Scale factor: $\frac{1}{2}$



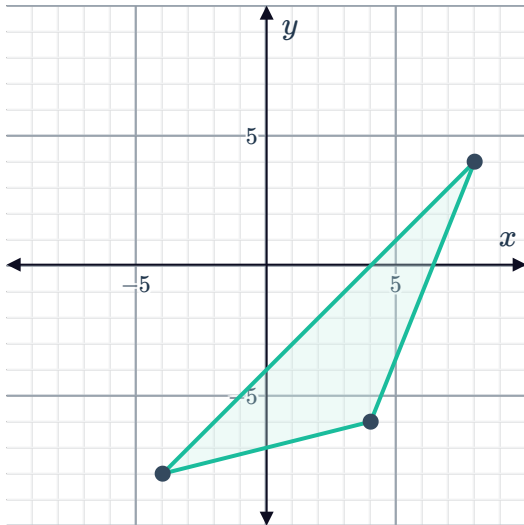
e Scale factor: $\frac{4}{5}$



Scale factor: $\frac{1}{2}$



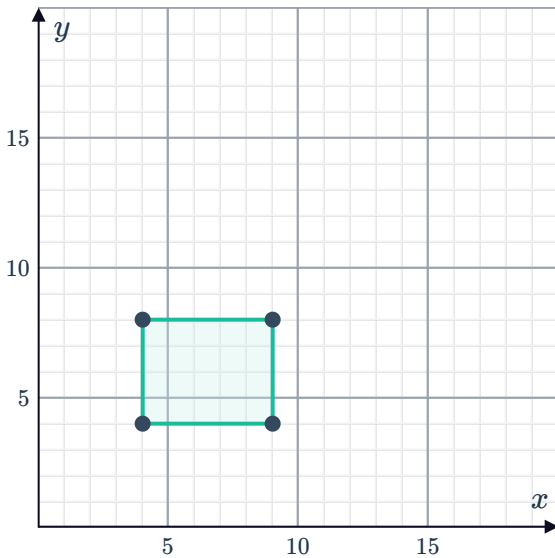
g Scale factor: $\frac{1}{2}$



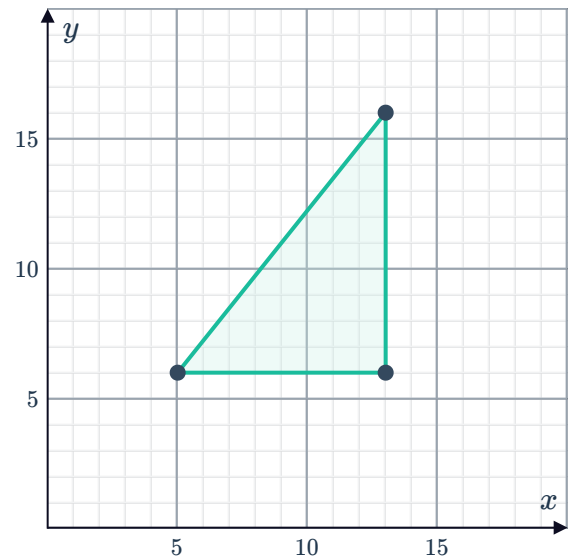
3 Dilate each figure using the given scale factor and center of dilation.



- a Scale factor: 2
Center of dilation: $A(2, 3)$

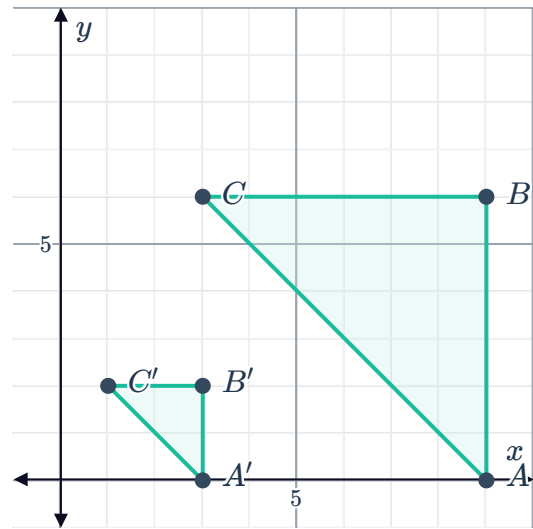


- b Scale factor: $\frac{1}{2}$
Center of dilation: $A(3, 2)$



- 4 A triangle with vertices $A(-7, -5)$, $B(3, -5)$ and $C(-5, 9)$ is dilated by a factor of 0.5 using the origin as the center of dilation. State the coordinates of the vertices of the dilated triangle.
- 5 If a square with an area of 25 ft^2 is dilated by a factor of 0.4, find the side length of the dilated square.
- 6 Find the scale factor of the following dilations:
- a A rectangle with vertices $A(-5, 3)$, $B(3, 3)$, $C(3, -5)$ and $D(-5, -5)$ is dilated using the origin as the center of dilation. The vertices of the new rectangle are $A'(-25, 15)$, $B'(15, 15)$, $C'(15, -25)$ and $D'(-25, -25)$.
- b A rectangle with vertices $A(-6, 9)$, $B(9, 9)$, $C(9, -12)$ and $D(-6, -12)$ is dilated using the origin as the center of dilation. The vertices of the new rectangle are $A'(-4, 6)$, $B'(6, 6)$, $C'(6, -8)$ and $D'(-4, -8)$.

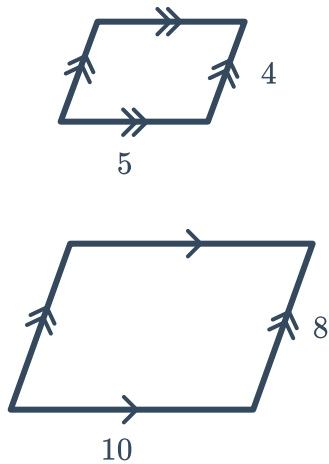
- 7 $\triangle ABC$ is dilated by a scale factor with a centre of dilation at the origin to obtain triangle $A'B'C'$.
Find the scale factor.



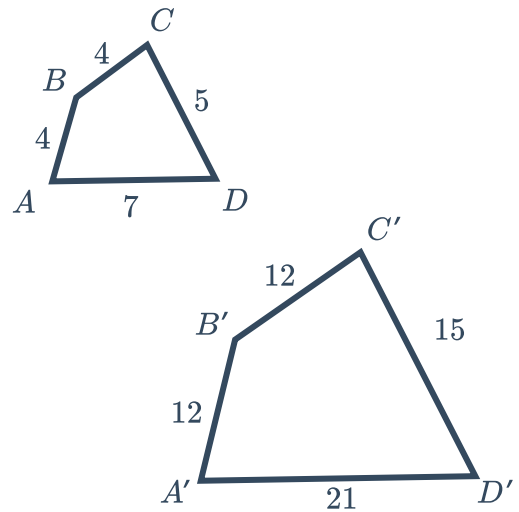
Additional questions

- 8 Find the scale factor for each pair of similar figures:

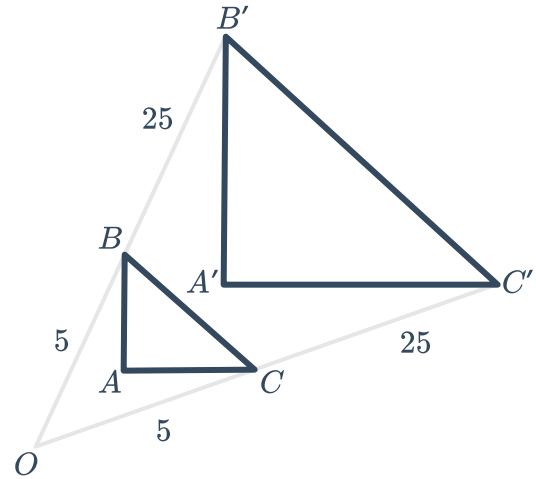
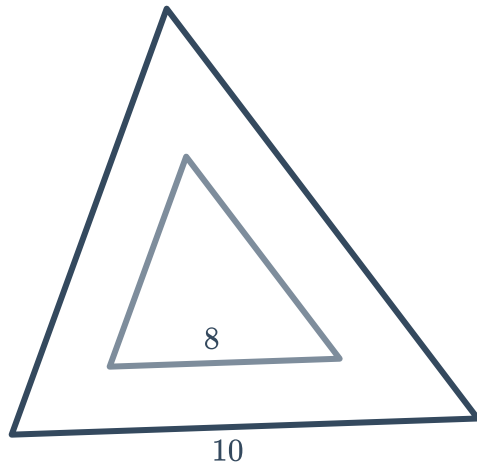
a



b



c



- 9 Determine whether each statement is true or false. Justify your answer.
- Any dilation with a scale factor less than 1 is a reduction.
 - Any dilation with a scale factor greater than 1 is an enlargement.
 - After dilating a figure, all corresponding angles will be congruent.
- 10 Layla describes the scale factor of a reduction as being in the form of $\frac{1}{k}$ since this is always a fraction less than one. Explain the error in her reasoning.
- 11 Use dilations to create a target: a bullseye made with four concentric circles where the width between rings is congruent. Create a diagram and justify your design.
- 12 Yousef needs to fit a **full** single sided sheet of notes on a single 3×5 -inch notecard for his science test. Yousef's original notes are written on a standard 8.5×11 -inch sheet of paper.
- Determine the largest scale factor Yousef can use to make sure the original notes fit on the notecard.
 - Find the dimensions of Yousef's notes after applying the scale factor.
 - Yousef realizes he can't read any of his notes because the font is smaller than $\frac{1}{12}$ of an inch after dilation. Describe how Yousef can divide and dilate his sheet of paper to utilize both sides of his notecard and still have readable notes.